CHE 321 1/26/26
Problem Set 1

Q1-2

What change cannot increase the rate constant (k)?

- can't answer with the current information I learned
- more of a chemistry question than an engineering one

Q1-5

the goal of the book is to teach reaction engineering through reasoning and understand the condition for each type of reactor. the website includes practice problems for each chapter and additional material to referance.

P1-1

1. $r = -kC_A C_B$

$$\frac{d(C_a)}{d(V)} = \frac{r_a}{v_0} \qquad \frac{d(C_b)}{dV} = \frac{r_b}{v_0}$$

$$\frac{r_a}{-k} = C_a C \qquad r_a = -k C_a \qquad r_b = -r_a$$

- ~ varying v_0 changes the $\frac{d(C)}{d(V)}$ which affect the change in C_A & C_B over the volume
- an increasing k decreases C_a and increases C_b

2. k - affects the speed of the reaction
v_0 - affects the conversion of the reactant
k_e - higher at higher v_0. lower at lower v_0

3. vertical thinking
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-

questions?

4) ?

P 1.5 $A + B \rightarrow 2C$ $r = -kC_A C_B$

$$\frac{dN_\alpha}{dt} = F_{\alpha,in} - F_{\alpha,out} + G_\alpha$$

$$G_\alpha = \int_V dV r_\alpha$$

$$\frac{dN_A}{dt} = F_{A0} - F_A + r_A V$$

$-F_C + \int^V r_C \, dV = \frac{dN_C}{dt}$

$$\frac{dN_B}{dt} = F_{B0} - F_B + r_B V$$

~~1.~~
~~2.~~
~~3.~~
~~4.~~
5.

$$\frac{dN_C}{dt} = -F_C + r_C V \rightarrow \frac{dN_C}{dt} = -F_C + \int_V dV r_C$$

$$3r_A' = r_C'$$

$?$ $$\frac{dN_C}{dt} = -F_C + 3\int_V dV r_A$$